When switch 'S' is open, there will be no gate current and the TRIAC is cut-off. Even with no gate current, the TRIAC can be turned 'ON' by providing the supply voltage equal to break over voltage (V_{BO}). However, the normal way to turn 'ON' TRIAC is by applying proper gate current.

When switch 'S' is closed, the gate current starts flowing in the circuit. The break over voltage of TRIAC can be varied by making proper gate current to flow by applying few mA at the gate. Now, the TRIAC will start conducting whether terminal MT_2 is positive or negative with respect to MT_1.

If the MT_2 is positive with respect to MT_1, the TRIAC turns *ON* and the conventional current will flow from MT_2 to MT_1. If the MT_2 is negative with respect to MT_1, the TRIAC is again turned *ON* but this time the conventional current flows from MT_1 to MT_2.

Application of TRIAC

1. It can be used as a static switch to turn AC power ON and OFF.
2. It is used for motor speed control.
3. It is used for illumination control.
4. It is used for heater control.
5. It is used for phase control.

6.14. INSULATED GATE BIPOLAR TRANSISTOR (IGBT)

IGBT is a three terminal semiconductor device with huge bipolar current carrying capability. So, this device is designed to make use of the benefits of both BJT and MOSFET devices in the form of monolithic.

IGBT has several applications in power electronics, particularly, PWM, UPS, SMPS and other power circuits. It increases the efficiency, dynamic performance and reduces the level of the audible noise. IGBT are also named as bipolar MOS transistor and conductivity modulated field effect transistor (COMFET).

History: The IGBT was proposed by Yamagami in 1968, but experimentally reported by B.W.Scharf and J.D.Plummer in 1978.

Construction: Figure 6.23 shows the structure, equivalent circuit and symbol of an IGBT. It is similar to the structure of MOSFET and the main difference

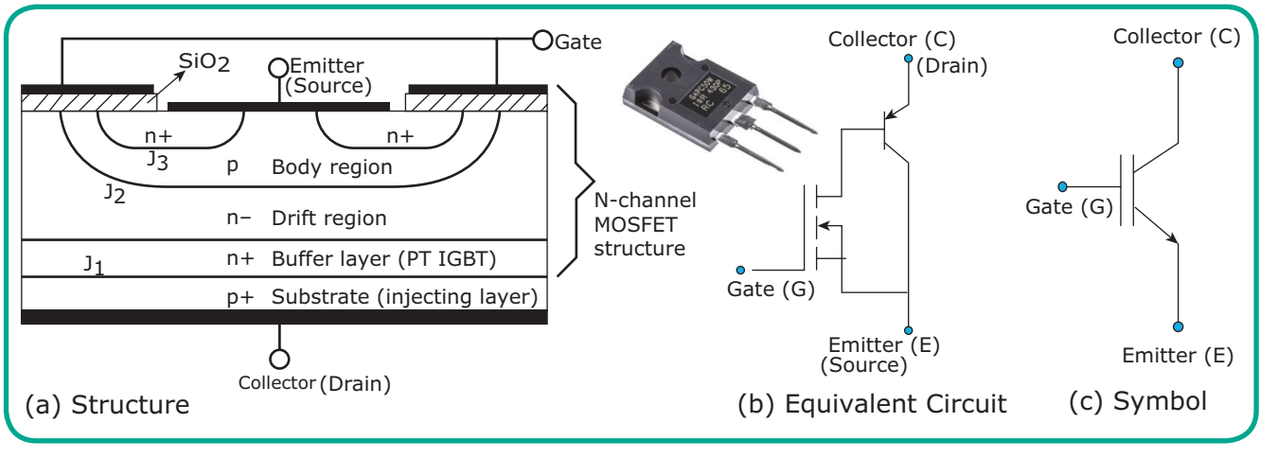


Figure 6.23 Insulated Gate Bipolar Transistor

ELEC_EQUIP_Chapter06.indd 142 11/10/2020 11:44:10 AM